

Digital Media - Midterm Exam

INSTRUCTIONS:

Overview and Objective

Your midterm task is to design, code and deploy an **Interactive Informative Web Page** on **one core Digital Media Topic** covered in class that interests you.

Your website must act as a functional, interactive educational manual or tool that explains the theoretical mechanics of your chosen topic to a user, while demonstrating professional web system standards (clean semantic HTML, CSS styling, responsive layout, and client-side JavaScript interactivity).

Topic Selection

Option A: Signal Digitization & Sampling

Option B: Digital Color Systems & Quantization

Option C: Raster VS Vector Graphics & Compression

Option D: Digital Typography

Mandatory Website Requirements

Module 1

Your web page must natively integrate and display **all listed elements** of digital media within your layout:

1. **Text:** Well-structured, readable instructional copy using semantic headings (`<h1>`–`<h3>`, `<p>`).
2. **Graphics:** Native SVG element (e.g., an interactive diagram, flowchart, or vector visualizer).
3. **Images:** Raster graphics (`.webp`, `.png`, or `.jpg`) demonstrating proper asset optimization.
4. **Audio:** Embedded native HTML5 `<audio>` player or Web Audio API synth node.
5. **Interactive Element:** A client-side JavaScript widget (e.g., a slider, side-by-side visual comparison, dynamic audio synthesizer, or live calculator). (Additional Point)

Module 2 Typography & Screen Legibility

- **Sans-Serif Primary Typeface:** Implement custom web fonts via CSS `font-family` optimized for screen legibility.
- **Micro-Typography CSS:** Explicitly configure micro-typographic CSS rules:
 - Leading: `line-height` (set between `1.4` and `1.6` for readable body text)
 - Tracking: `letter-spacing` (fine-tuned for headers/labels)
 - Kerning: `font-kerning: normal;`

Module 3 Technical Media Spec

At the bottom of your web page, you must include a **Technical Media Spec Table** auditing at least 3 embedded media assets used on your site:

Asset Name & File Format	Original Size → Compressed	Color Depth / Quantization	Compression Algorithm
e.g., <i>demo-audio.mp3</i>	3.2 MB to 480 KB	16-bit	Variable Length (Huffman)
e.g., <i>sample-visual.png</i>	1.8 MB to 210 KB	8-bit quantized (256 LUT)	Lossless (RLE)

Submission Phases:

Phase 1: Website Wireframe and Chosen Topic (September 28, 2026)

For this requirement, you may submit your wireframe in any format.

See the submission link: [M1. Phase 1: Digital Media Web Manual \(Midterm Exam\)](#)

Phase 2: Website Files Submission (October 8, 2026)

- **Source Code Archive (.zip):** Containing your `index.html`, `style.css`, and organized `/assets` sub-folder (`/images`, `/audio`, `/video`).
- **Live URL:** Link to your deployed site (e.g., GitHub Pages, Netlify Drop or Vercel) or just the link.

Phase 3: Individual Presentation (October 8 & October 12)

Your voice matters! To further understand how your work was crafted, we will have a scheduled presentation of each individual's work.

Discuss your accomplishments and be ready to answer inquiries and questions from me.